

Student
James Stuart

Total Points
100 / 100 pts

Question 1

(no title)

15 / 15 pts

✓ - 0 pts Correct

Question 2

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt Mass flow rate should be around 4.9 kg/s.
- 1 pt Area should be around 0.07 sq. m

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt Power should be around 3500 kW.

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 3 pts Incomplete
- 1 pt Heat transfer rate will be around -15 kW.

Question 5

(no title)

20 / 20 pts

✓ - 0 pts Correct

- 1 pt POWER=0.52kW
- 4 pts Incomplete

Question 6

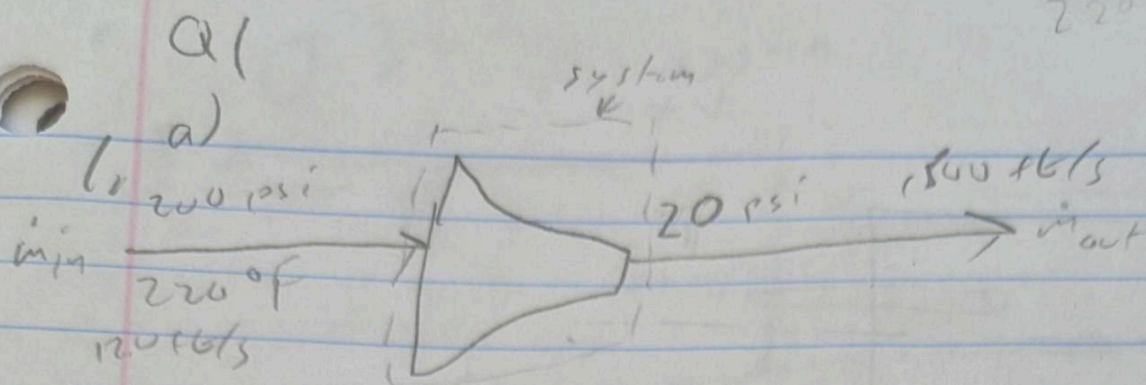
(no title)

20 / 20 pts

✓ - 0 pts Correct

- 1 pt x=0.191
- 6 pts inomplete

No questions assigned to the following page.



b) steady state
control volume
 $\Delta PE = 0$

c) M.B. $\dot{m}_{in} = \dot{m}_{out}$

E.B.

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \dot{m}(h_1 - h_2 + \frac{v_1^2}{2} - \frac{v_2^2}{2} + g(z_1 - z_2))$$

$$\dot{m}(h_1 - h_2 + \frac{v_1^2}{2} - \frac{v_2^2}{2}) = 0$$

$$h_2 - h_1 = \frac{v_1^2}{2} - \frac{v_2^2}{2}$$

d)

$$h_1 = 144.15 \frac{\text{Btu}}{\text{lb}} \rightarrow \text{table}$$

$$h_2 = \frac{1}{2} \left(\frac{120 \text{ ft/s}}{5} \right)^2 - \frac{1}{2} \left(\frac{1500 \text{ ft/s}}{5} \right)^2 + 144.15 \frac{\text{Btu}}{\text{lb}}$$

$$h_2 = 99.54 \frac{\text{Btu}}{\text{lb}}$$

$$v_1 = \frac{120 \text{ ft/s}}{5} = 0.5747 \frac{\text{ft/s}}{\text{lb}}$$

$$v_2 = \frac{1500 \text{ ft/s}}{5} = 89.79 \frac{\text{ft/s}}{\text{lb}}$$

$$T_2 = -2.48^\circ \text{F}$$

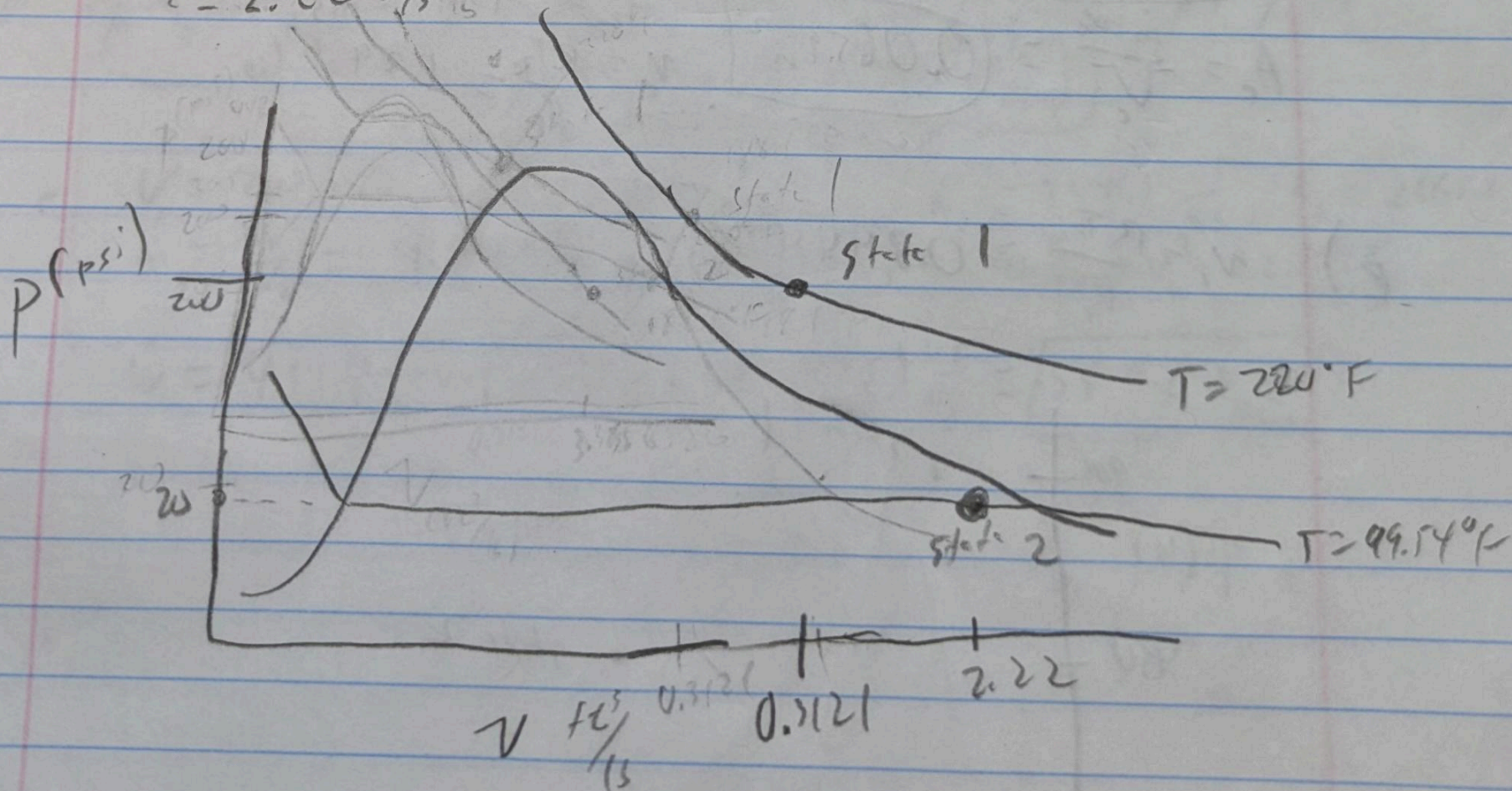
liquid vapor mix
 $x = 0.98$

$$\frac{f \text{ ft/s}}{5^2} \times \frac{14.5 \text{ ft/s}}{32.2 \text{ ft/s}^2} \times \frac{10 \text{ ft/s}}{178.1 \text{ ft/s}}$$

e)

$$v_1 = 0.3121 \frac{\text{ft}^3}{\text{lb}}$$

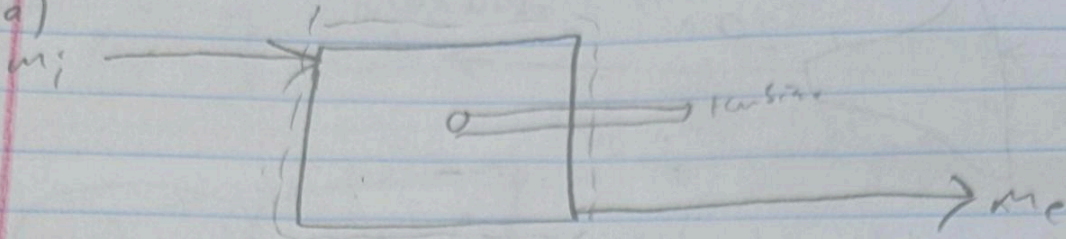
$$v_2 = 2.22 \frac{\text{ft}^3}{\text{lb}}$$



No questions assigned to the following page.

Q2

2. a)



b) steady state ideal gas

control volume

$$\Delta PE = 0$$

$$V_i = \text{"swell"} = 0$$

$$\dot{Q} = 0$$

c) $\dot{m}_i = \dot{m}_e$, $\frac{dE}{dt} = \dot{Q} - \dot{W} + \dot{m} \left(h_1 - h_2 + \frac{V_1^2}{2} - \frac{V_2^2}{2} + g z_1 - g z_2 \right)$

$$\dot{W} = \dot{m} \left(h_1 - h_2 - \frac{V_2^2}{2} \right)$$

d) $\dot{m} = \frac{2500 \frac{kg}{s}}{(1000 - 451.8 - 4.65) \frac{kg}{kg}}$

$$\dot{m} = 4.59 \frac{kg}{s}$$

$$A_c = \frac{\dot{m} V_c}{V_c} = 0.0659 m^2$$

$$\frac{\dot{Q}}{\dot{m}} = \frac{1.2}{5}$$

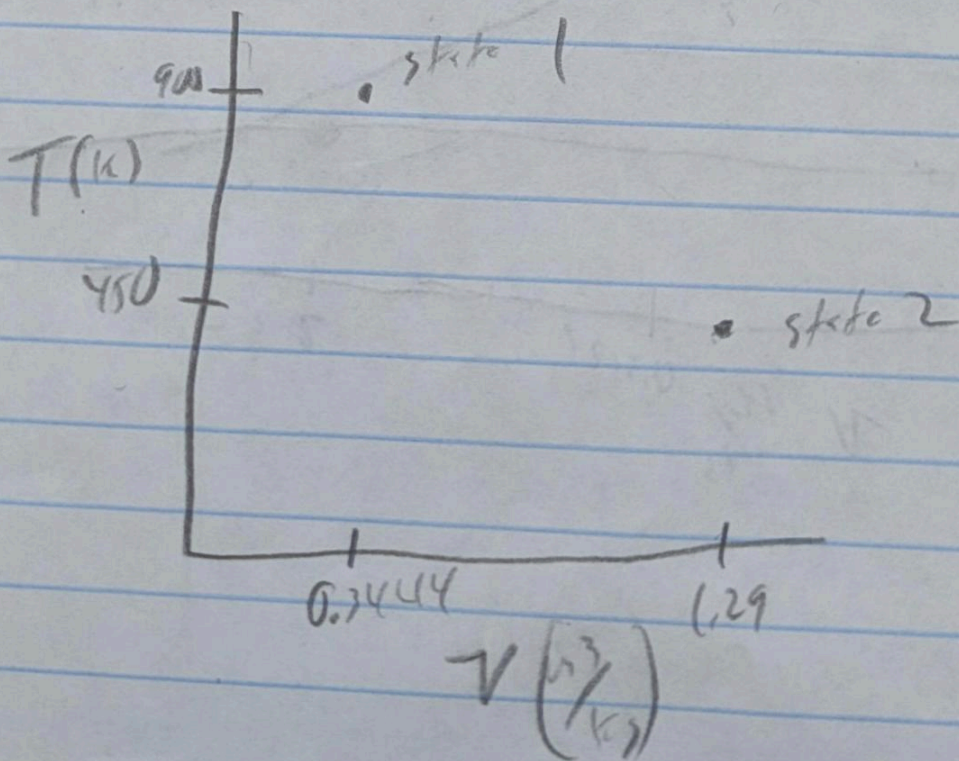
$$V_c^2 = 8100 \frac{m^2}{s^2} = 8.1 \frac{kJ}{kg}$$

$$h_1 = 1600.55 \frac{kJ}{kg}$$

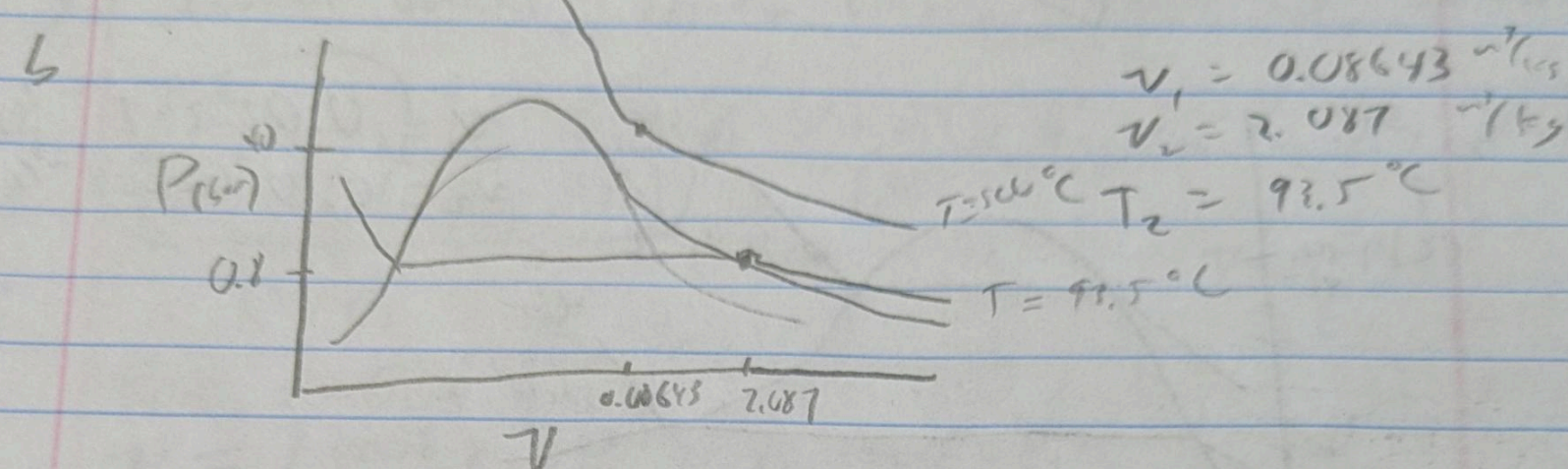
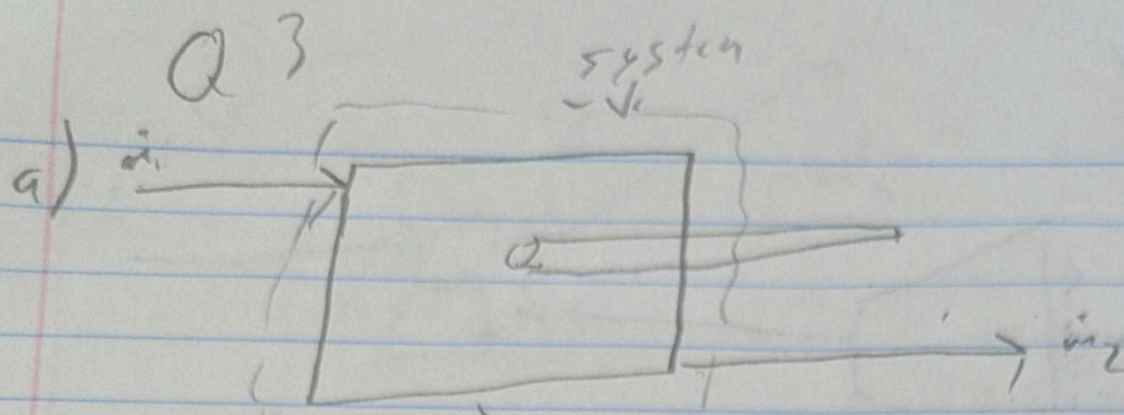
$$h_2 = 451.80 \frac{kJ}{kg}$$

$$V_c = \frac{RT_c}{P_c} = 1.29 \frac{m^3}{kg}$$

e) $v_i = \frac{RT_i}{P_i} = 0.3444 \frac{m^3}{kg}$



No questions assigned to the following page.



c. control volume
steady state
Adiabatic
 $\Delta PE = 0$

d. m^3

$m_1 = m_2$

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \dot{m}(h_1 - h_2 + \frac{V_1^2}{2} - \frac{V_2^2}{2} + g z_1 - g z_2)$$

EB $\boxed{\dot{W} = \dot{m}(h_1 - h_2 + \frac{V_1^2}{2} - \frac{V_2^2}{2})}$

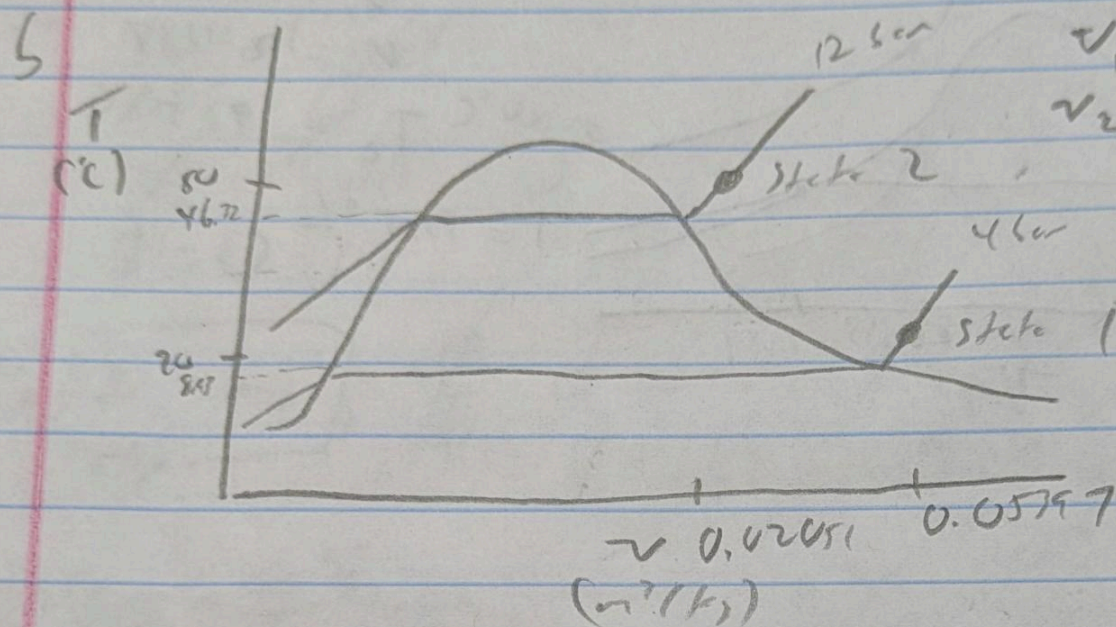
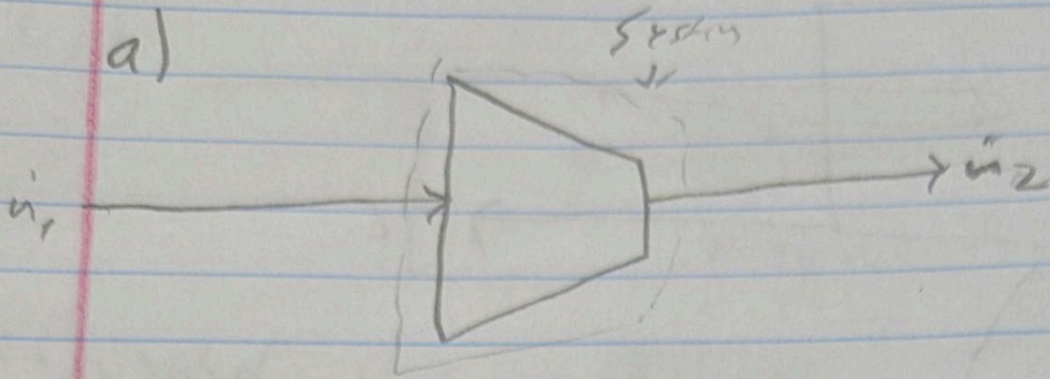
e

$V = 200 \text{ m/s}$ $V_2 = 150 \text{ m/s}$ $h_1 = 3445.3 \frac{\text{kJ}}{\text{kg}}$ $h_2 = 2665.8 \frac{\text{kJ}}{\text{kg}}$
 $\dot{m} = \frac{\dot{V}_2}{v_2} = 4.54 \text{ kg/s}$ $V_1^2 = 40000 \frac{\text{m}^2}{\text{s}^2} = 40 \frac{\text{kJ}}{\text{kg}}$ $V_2^2 = 22.5 \frac{\text{kJ}}{\text{kg}}$

$\dot{W} = 4.54 \frac{\text{kg}}{\text{s}} (3445 - 2665.8 + 20 - 11.25) \frac{\text{kJ}}{\text{kg}} = \boxed{3579 \text{ kW}}$

No questions assigned to the following page.

Q4



c. Control volume
steady state
 $\Delta KE = \Delta PE = 0$

$$\dot{V}_1 = \frac{4 \text{ m}^3}{\text{s}} = 0.0667 \text{ m}^3/\text{s}$$

d. $\dot{m}_1 = \dot{m}_2$ $\dot{W} = \dot{Q} + \dot{m}(h_1 - h_2)$

e. $\dot{Q} = \dot{W} - \dot{m}(h_1 - h_2)$

$$\dot{W} = \left(-60 \frac{\text{kJ}}{\text{kg}}\right) \left(\frac{1.235 \text{ kg/s}}{1}\right) = -74.1 \frac{\text{kJ}}{\text{s}}$$

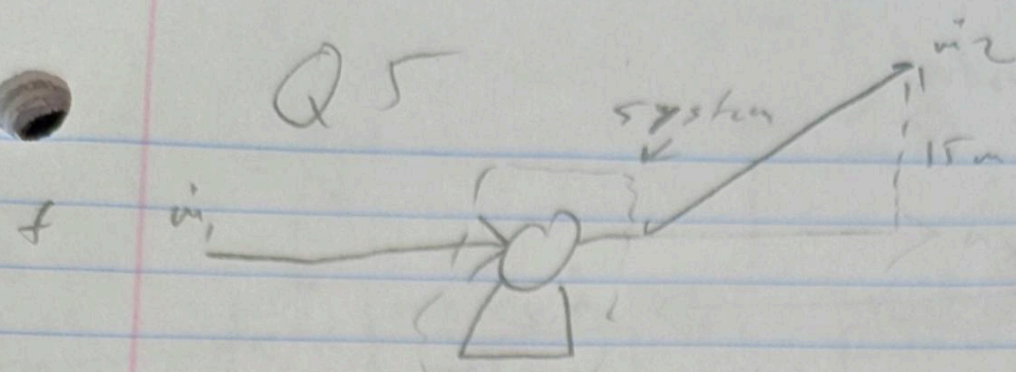
$$h_1 = 262.96 \frac{\text{kJ}}{\text{kg}}$$

$$h_2 = 310.24 \frac{\text{kJ}}{\text{kg}}$$

$$\dot{m} = \frac{\dot{V}_1}{v_1} = 1.235 \frac{\text{kg}}{\text{s}}$$

$$\dot{Q} = -74.1 \text{ kW} - 1.235 \frac{\text{kg}}{\text{s}} (262.96 - 310.24) \frac{\text{kJ}}{\text{kg}} = \boxed{-15.7 \text{ kW}}$$

No questions assigned to the following page.



g Control volume, steady state, $\Delta KE = 0$, adiabatic

h $\dot{m}_1 = \dot{m}_2$ $\frac{dE}{dt} = \dot{Q} - \dot{W} + \dot{m} \left(h_1 - h_2 + \frac{V_1^2}{2} - \frac{V_2^2}{2} + g z_1 - g z_2 \right)$
 $\dot{W} = \dot{m} (h_1 - h_2 + g z_1 - g z_2)$ $\dot{W} = (\dot{m} / g) (g z_1 - g z_2)$
 $h_1 - h_2 = v (p_1 - p_2)$

i $\dot{m} = 1.5 \text{ kg/s}$ $z_1 - z_2 = -15 \text{ m}$

$h_1 - h_2 = -0.2 \frac{\text{kJ}}{\text{kg}} (-15 \text{ m}) = -200 \text{ J/kg}$ $\dot{W} = 0.22 \text{ kW}$

$v = 1.0007 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}$ $p_1 - p_2 = -200 \text{ kPa}$

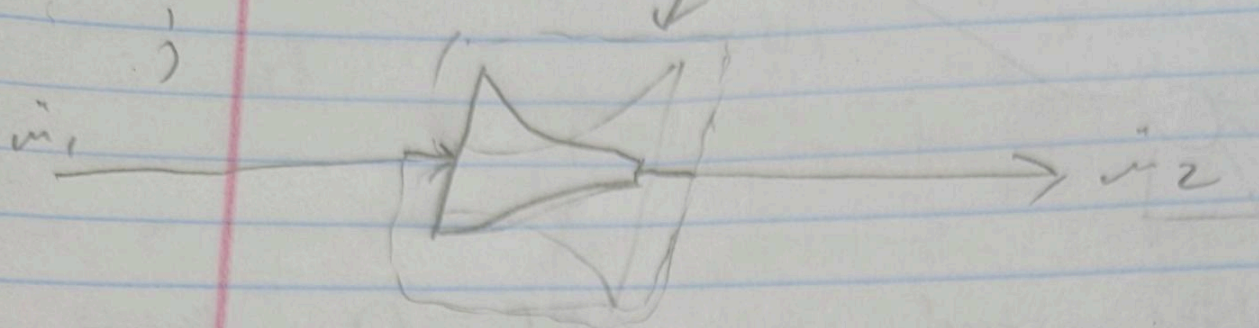
$\dot{W} = 1.5 \text{ kg/s} \left(-200 \frac{\text{J}}{\text{kg}} + \left(\frac{9.8 \text{ m}}{\text{s}^2} \right) (-15 \text{ m}) \right) = -0.52$

0.52 kW required to run the pump

No questions assigned to the following page.

Q6

system



K control volume, steady state, $\Delta PE = 0$, $W = 0$

L $\dot{m}_1 = \dot{m}_2$ $\frac{dE}{dt} = \dot{Q} - \dot{W} + \dot{m}_1 \left(h_1 + \frac{V_1^2}{2} + gz_1 \right) - \dot{m}_2 \left(h_2 + \frac{V_2^2}{2} + gz_2 \right)$

Q $EP_2 = (h_1 = h_2)$

$h_1 = 297.76$

M $x = 0.197$

$v_1 = 1.6504 \times 10^{-3} \text{ m}^3/\text{kg}$

$v_2 = 0.2254 \text{ m}^3/\text{kg}$

